

REVIEW - SYSTEMATIC

Effects of phytoestrogen supplementation on intermediate cardiovascular disease risk factors among postmenopausal women: a meta-analysis of randomized controlled trials

Maike Wolters, PhD,¹ Gordana M. Dejanovic, MD,^{2*} Eralda Asllanaj, MD, DSc,^{3,4*} Kathrin Günther, PhD,¹ Hermann Pohlabein, PhD,¹ Wichor M. Bramer, MSc,⁵ Jenny Ahrens, BA,¹ Rajini Nagrani, PhD,¹ Iris Pigeot, PhD,¹ Oscar H. Franco, MD, PhD,⁶ Wolfgang Ahrens, PhD,¹ Taulant Muka, MD, PhD,⁶ and Marija Glisic, MD, PhD^{1,6,7}

Abstract

Importance: Phytoestrogens are becoming popular constituents of human diets and are increasingly used by postmenopausal women.

Objective: Our study aims to determine the effects of phytoestrogen supplementation on intermediate cardiovascular disease (CVD) risk factors in postmenopausal women.

Evidence review: Five electronic databases (Medline, EMBASE, Web of Science, Cochrane CENTRAL, Google Scholar) were systematically searched to identify eligible studies, that is, randomized controlled trials (RCTs) that assessed the association of phytoestrogen supplementation with CVD risk factors (serum lipids, homocysteine, fibrinogen, markers of inflammation, oxidative stress and endothelial function, carotid intima-media thickness [CIMT]) in postmenopausal women. Data were extracted by two independent reviewers using a predefined data collection form.

Findings: In total, 56 RCTs were identified, including 4,039 individual postmenopausal women. There was substantial heterogeneity in quality across studies. Twenty-six (46%) RCTs showed poor quality and there was an indication of publication bias presence for some of the biomarkers. Results are reported in pooled mean difference (95% CI) of changes. Use of phytoestrogens was associated with a decrease in serum total cholesterol (−0.27 mmol/L [−0.41 to −0.13]), low-density lipoprotein (−0.25 mmol/L [−0.37 to −0.13]), triglycerides (−0.20 mmol/L [−0.28 to −0.11]), and apolipoprotein B (−0.13 g/L [−0.23 to −0.03]) and with an increase in serum apolipoprotein A-1 (0.04 g/L [0.02-0.07]). Also, phytoestrogen supplementation was associated with a decrease in serum intercellular adhesion molecule 1 (−18.86 ng/mL [−30.06 to −7.65]) and E-selectin (−2.32 ng/mL [−4.05 to −0.59]). There was no association observed between phytoestrogen supplementation and inflammatory markers, fibrinogen, homocysteine, or other endothelial function markers. In contrast, use of phytoestrogens was associated with an increase in CIMT (9.34 μ m [95% CI, 0.39-18.29]). Effect estimates of phytoestrogen supplementation on oxidative stress could not be pooled.

Conclusions and Relevance: Phytoestrogen supplementation seems to modestly improve the CVD risk profile of postmenopausal women by influencing blood lipids and parameters of endothelial function. In women with an increased risk of atherosclerosis, although modest, a harmful effect on CIMT progression may be present. Because of limited quality and the heterogeneous nature of the current evidence, additional rigorous studies are needed to explore the role of phytoestrogens in menopausal cardiovascular health.

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From the ¹Leibniz Institute for Prevention Research and Epidemiology - BIPS, Bremen, Germany; ²Department of Ophthalmology, University of Novi Sad, Faculty of Medicine, Novi Sad, Serbia; ³Department of Epidemiology, Erasmus MC, University Medical Center Rotterdam, Rotterdam, The Netherlands; ⁴Institute for Community Medicine, University of Greifswald, Greifswald, Germany; ⁵Medical Library, Erasmus MC, University Medical Center Rotterdam, Rotterdam, The Netherlands; ⁶Institute of Social and Preventive Medicine (ISPM), University of Bern, Bern, Switzerland; and ⁷Swiss Paraplegic Research, Nottwil, Switzerland.

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*G.M.D. and E.A. denote equal contribution.

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Address correspondence to: Maike Wolters, PhD, Department of Epidemiological Methods and Etiological Research, Leibniz Institute for Prevention Research and Epidemiology - BIPS, Achterstr. 30, 28359 Bremen, Germany. E-mail: wolters@leibniz-bips.de

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Increasing numbers of women use phytoestrogen supplements.^{1,2} Phytoestrogens, plant-derived estrogen-like compounds, may cause organ-specific (anti-)estrogenic effects and may modify estrogen-dependent signaling pathways.^{2,3} High levels of estrogen in postmenopausal women, as well as use of menopausal hormone therapy, were associated with adverse changes in cardiometabolic health (including CVD),⁴⁻⁷ raising a concern regarding potential cardiovascular consequences of phytoestrogens in aging women.^{2,3}

Although favorable cardiovascular health effects have been ascribed to phytoestrogens by positively altering the levels of cardiovascular risk factors such as lipids, blood pressure, and inflammation and atherosclerosis,⁸⁻¹⁰ the evidence has been inconsistent. Various meta-analyses have attempted to summarize the evidence from randomized controlled trials (RCTs) to quantify the association between phytoestrogen supplementation and CVD risk factors. The majority of previous meta-analyses were, however, limited by (i) focusing on heterogeneous interventions (eg, phytoestrogens plus exercise)¹¹ or solely one outcome¹²; (ii) including specific types of phytoestrogens: isoflavones,¹³ flaxseed,¹⁴ or soy products only¹⁵⁻¹⁷; and (iii) inclusion of heterogeneous populations in terms of menopausal status or sex^{12,16,17} making interpretation of results challenging.

Therefore, we conducted a systematic review and meta-analysis of RCTs evaluating the association of phytoestrogen supplementation with CVD risk factors in postmenopausal women.

METHODS

Data sources and search strategy

The current review was conducted following a standardized protocol registered in PROSPERO (ID No. CRD42019121110) and in accordance with the PRISMA Statement¹⁸ and the *Cochrane Handbook for Systematic Reviews of Interventions*.¹⁹ An experienced information specialist (W.M.B.) searched four bibliographic databases: Medline ALL via Ovid (from 1946), EMBASE via embase.com (from 1974), Web of Science Core Collection (from 1900), Cochrane CENTRAL via Wiley (from 1996) from inception until January 23, 2020 (date last searched). In addition, we downloaded the first 200 results from the search engine Google Scholar. The searches combined terms related to (i) supplementation such as “phytoestrogen,” “red clover,” “soybean,” etc and (ii) CVD risk factors (eg, lipoproteins, inflammation markers) and were filtered to include human studies only. Furthermore, the reference lists of the included studies and relevant reviews were searched for eligible studies. Details on the search strategy are provided in Supplementary Table 1, <http://links.lww.com/MENO/A627>.

KEY POINTS

Question: Does phytoestrogen use improve cardiovascular disease risk factors in postmenopausal women?

Objective: The aim of the study was to determine the effect of phytoestrogen supplementation on CVD risk factors by means of a systematic review and meta-analysis of randomized controlled trials.

Findings:

- This meta-analysis shows that phytoestrogen supplementation improves serum triglycerides, total cholesterol, low-density lipoprotein, apolipoproteins A-1 and B as well as cell adhesion molecules (intercellular adhesion molecule 1, E-selectin) but not other endothelial function markers, flow-mediated diameter, fibrinogen, or inflammation markers (high quality of evidence).
- Phytoestrogens have no effect on homocysteine (moderate quality of evidence).
- Use of phytoestrogens may have a modest adverse effect on carotid intima-media thickness progression particularly in postmenopausal women at increased risk of developing atherosclerosis (low quality of evidence).

Meaning: Phytoestrogen supplementation may be useful to improve the cardiovascular disease (CVD) risk profile in postmenopausal women but caution is indicated in women with an increased risk of developing atherosclerosis.

Study selection, eligibility criteria, and data extraction

Detailed inclusion and exclusion criteria can be found in the review protocol (PROSPERO ID No. CRD42019121110). In brief, RCTs were eligible for inclusion if they (i) were conducted among postmenopausal women and (ii) investigated associations of phytoestrogen supplementation with any of the following outcomes: serum lipids, inflammatory markers, coagulation system/fibrinolysis markers, homocysteine, markers of oxidative stress, markers of endothelial dysfunction, and vascular function and carotid atherosclerosis. Exclusion criteria were (i) RCTs investigating phytoestrogen supplementation in combination with dietary restrictions or other interventions, (ii) inappropriate study population (premenopausal women or men), and (iii) articles with incomplete information.

Based on these criteria, titles and abstracts were independently evaluated by two reviewers. The full texts of potentially eligible studies were afterwards assessed by two

independent reviewers and any disagreement was settled by reaching a consensus or by consulting a third reviewer. Two authors independently extracted the relevant information using a predefined data extraction form.

Assessing the quality of evidence

The quality of included RCTs was assessed by two independent reviewers using the Cochrane Collaboration's tool.¹⁹ Detailed information on the assessment of study quality and risk of bias is provided in Supplementary Table 2, <http://links.lww.com/MENO/A627>. Furthermore, we applied the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach to rate the overall quality of the evidence for risk of bias, publication bias, imprecision, inconsistency, indirectness, and magnitude of effect.²⁰

Data synthesis and statistical analysis

Treatment effects were defined as the pre-post differences in outcomes between phytoestrogen supplementation and control at the end of the trial. All outcomes were continuous; therefore, the mean differences (intervention minus control) of the treatment effects in CVD risk factors were presented as summary outcome measures. For data reported as medians, ranges, or 95% confidence intervals (CIs), we calculated means and standard deviations using previously described method.²¹ Units of measurement were converted to SI units where appropriate. Random-effect models were used to obtain estimates of weighted mean differences and 95% CIs. For RCTs with crossover design we used the data from the first period only. Fixed effect models as shown in the forest plots were applied to report the estimates for sensitivity analyses. Heterogeneity was assessed using the Cochrane χ^2 statistic and the I^2 statistic and was determined as low ($I^2 \leq 25\%$), moderate ($25\% < I^2 < 75\%$), or high ($I^2 \geq 75\%$).²²

Study characteristics including geographic location, number of participants, duration of intervention, type of supplement's administration, baseline age and years since menopause onset, body mass index, smoking status, women's health status, hyperlipidemia, and study quality were prespecified as characteristics for assessment of heterogeneity and were evaluated using stratified analyses and random-effects meta-regression if eight or more studies were included in the meta-analysis.²³ We performed a leave-one-out sensitivity analysis iteratively by removing one study at a time to confirm that the findings were not influenced by any single study. Asymmetry was assessed by Egger's test and publication bias was evaluated through a funnel plot. All statistical analyses were conducted with STATA, Release 14 (Stata Corp, College Station, TX). The trials that could not be quantitatively pooled were descriptively summarized.

RESULTS

Study identification and selection

Based on the search strategy, 13,780 citations were identified of which 142 were selected for detailed full text evaluation. Of those, 65 studies based on 56 unique RCTs met the

inclusion criteria and were included in our systematic review (Fig. 1). Among these (i) 42 RCTs investigated serum lipids (total cholesterol [TC], low-density lipoprotein [LDL] cholesterol, high-density lipoprotein cholesterol [HDL], triglycerides [TGs], apolipoprotein A-1 and B [Apo A-1/B], lipoprotein a); (ii) 10 RCTs investigated inflammatory markers (C-reactive protein [CRP], interleukin-6 [IL-6], tumor necrosis factor α [TNF- α]); (iii) four hemostatic factors (fibrinogen); (iv) five homocysteine; (v) seven endothelial metabolites and eight cell adhesion molecules (intercellular adhesion molecule 1 [ICAM-1], vascular cell adhesion molecule 1 [VCAM-1], E-selectin, P-selectin, nitric oxide products (NOx), endothelin-1); (vi) four vascular function (flow-mediated diameter [FMD]); (vii) three studied carotid intima-media thickness (CIMT) (Table 1, Supplementary Table 3, <http://links.lww.com/MENO/A627>) and seven RCTs investigated the association of phytoestrogen supplementation and oxidative stress (Supplementary Table 4, <http://links.lww.com/MENO/A627>).

Characteristics of included studies

The review included 4,039 postmenopausal women from 56 RCTs. Seventeen RCTs were conducted in North America, 12 in Europe, 15 in Asia-Pacific, 9 in South America, and 9 in Middle East. The sample size ranged from 16 to 325 women (median 50.5 women) and the duration of the interventions from 2 to 113 weeks (median duration 12 wk). The majority of RCTs included healthy women ($n=41$), whereas the rest of RCTs recruited women with different health conditions (eg, type 2 diabetes [T2D]). Most of the RCTs ($n=44$) investigated dietary soy products or isoflavones from soy, four trials reported on interventions with red clover and one study examined a combination of soy extract with black cohosh extract. Five trials examined isolated isoflavones without specifying the source, two studied flaxseed lignans and one trial studied lignans from sesame (Supplementary Table 5, <http://links.lww.com/MENO/A627>).

Inflammation markers, serum lipids, fibrinogen, and homocysteine

Based on findings from 40 RCTs²⁴⁻⁶⁵ including 3,069 women, phytoestrogen supplementation, compared to placebo, was associated with a moderate decrease in TC, LDL, and TG with pooled mean differences of changes of -0.27 mmol/L (95% CI: -0.41 to -0.13), -0.25 mmol/L (-0.37 to -0.13), and -0.20 mmol/L (95% CI: -0.28 to -0.11), respectively. Based on 12 RCTs^{25,31,36,41,46,49,50,57,61,62,65,66} including 985 women, serum Apo A-1 increased (0.04 mmol/L [95% CI: 0.02 - 0.07]), whereas Apo B decreased (-0.13 mmol/L [95% CI: -0.22 to -0.03]), in the phytoestrogen arm as compared to controls. There were no significant associations between phytoestrogen supplementation and serum Lp(a) or HDL. Also, no statistically significant associations were observed between phytoestrogen supplementation and serum CRP,^{24,25,44,66-72} TNF- α ,^{70,73-75} IL-6,^{70,73,74} fibrinogen,^{25,30,66,76} or homocysteine^{56,67,68,77,78} (Fig. 2, Supplementary Figs. 1-6, <http://links.lww.com/MENO/A627>).

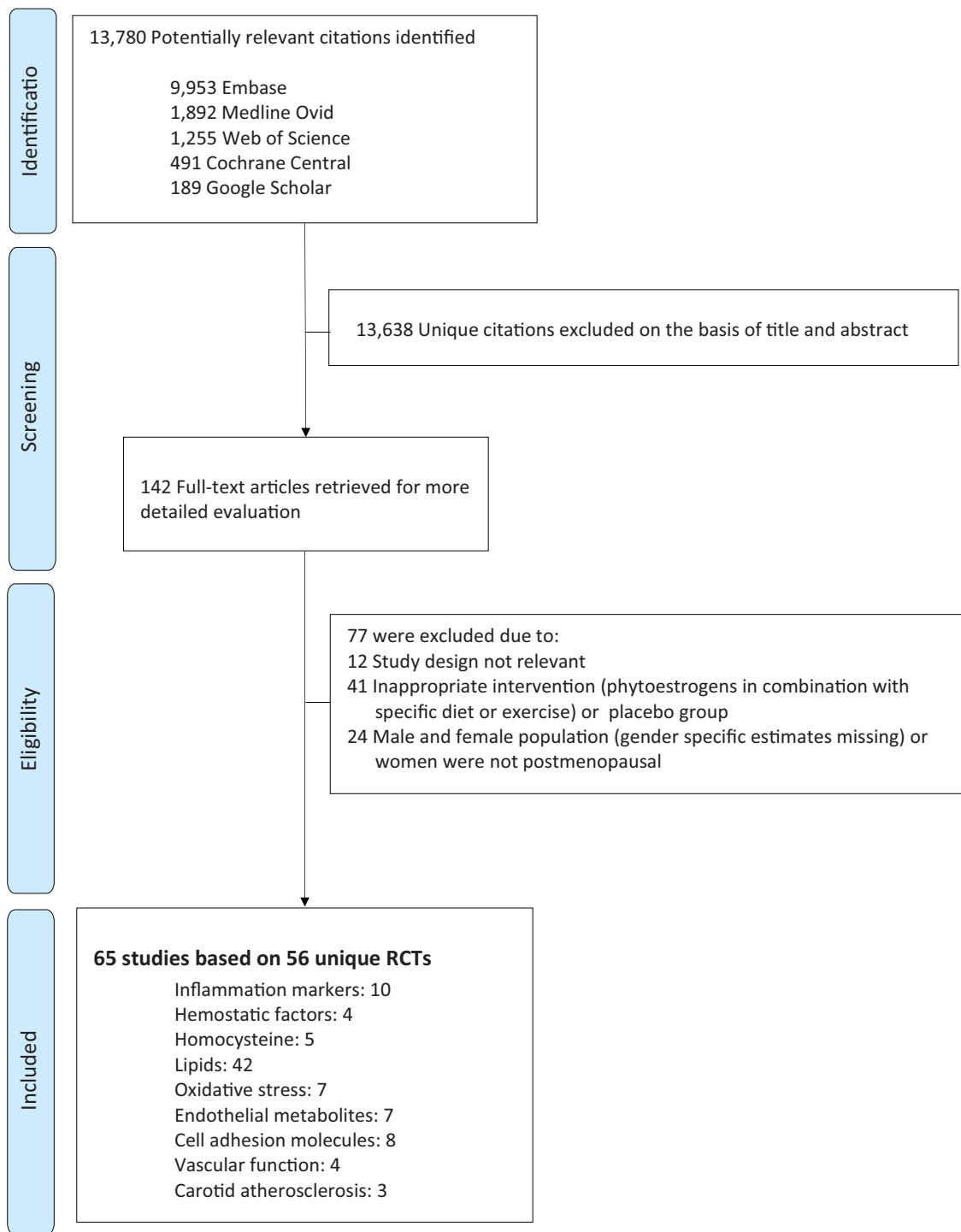


FIG. 1. Flowchart of randomized controlled trials included in the current review. RCT, randomized controlled trial.

Cell adhesion molecules, endothelial metabolites, vascular function, and carotid atherosclerosis

Based on findings from 5 RCTs including 583^{30,68-70,79} and 651^{30,68,69,72,79} women, respectively, phytoestrogen supplementation was associated with a decrease in serum ICAM-1 (-18.86 ng/mL [95% CI: -30.06 to -7.65]) and E-selectin (-2.32 ng/mL [95%CI: -4.05 to -0.59]) as compared to

placebo. Although no associations were observed between phytoestrogen supplementation and the other endothelial function markers: VCAM-1, endothelin-1,^{69,80} and NOx.^{72,80} Furthermore, based on three unique RCTs^{44,81,82} including 688 women, use of phytoestrogens was associated with a modest increase in CIMT (9.34 μ m/y [95% CI: 0.39 - 18.29]), whereas no association with FMD^{42,83-85} was observed

TABLE 1. Characteristics of the unique randomized controlled trials included in the meta-analyses

Outcome	Eligible studies			Participants			Location			Intervention characteristics						
	Unique studies, no	Follow-up duration, median (IQR), wk	Total	Median (IQR), no.	Age, median (IQR), y	Europe	North America	South America	Asia-Pacific	Administration type		Source of phytoestrogens				
										Pills/ tablets	Dietary products	Isoflavones (soy or isolated)	Coumestans (red clover)	Lignans (flax seed)		
Inflammation markers	CRP	10	14 (12-24)	1,077	90 (50-179.75)	57.5 (55.7-64.35)	5	3	—	1	1	4	6	8	—	2
	IL-6	3	4 (NA)	139	44 (NA)	54.61 (NA)	1	2	—	—	—	—	3	2	—	1
	TNF-α	4	9 (14.5-15)	163	38 (31.75-6)	56.2 (54.75-59.9)	1	3	—	—	—	—	4	3	—	1
	Fibrinogen	4	24 (24-42)	371	58.5 (51.8-149.3)	55.33 (54.8-62.1)	2	1	—	1	—	3	1	3	—	1
Hemostatic factors	Homocysteine	5	16 (7-24)	423	68 (49-128.5)	53 (NA) ^a	2	1	1	—	1	3	2	5	—	—
	TC	40	12 (12-24)	3,069	50 (38-90)	55.7 (53.9-58)	5	11	6	12	6	25	24	34	3	3
	TG	39	12 (12-24)	3,038	52.5 (37-90)	55.7 (53.9-57.9)	4	13	6	11	5	28	22	35	3	3
	Apo A-1/ Apo B	12	12 (12-24)	985	50 (36-90)	53.9 (52.6-56.9)	2	2	2	5	1	8	8	11	0	1
Cell adhesion molecules	LP (a)	7	12 (12-24)	832	98 (41-162)	54.9 (52.1-56.9)	3	2	1	1	—	4	4	6	—	1
	VCAM-1/ ICAM-1	5	8 (6-24)	583	94 (53.8-148.5)	56.12 (55.42-59.45)	3	1	—	1	—	2	3	4	—	1
	E-selectin	5	12 (7-24)	651	116.7 (62.3-148.5)	55 (54.58-56.7)	3	1	—	1	—	3	2	5	—	—
	NO products	6	10 (7.5-30)	573	91.5 (53-128.5)	57 (52.75-61.75)	5	—	—	—	1	4	2	4	—	2
Endothelial metabolites	Endothelin-1	4	7 (6-38)	427	75.5 (48-198.7)	59.4 (57.12-62.13)	4	—	—	—	—	—	4	2	—	2
	FMD	4	6 (4.5-19.5)	442	112 (41-180)	60 (58-61.6)	—	3	—	1	—	2	2	4	—	—
Vascular function	CIMT	3	48 (NA)	688	180 (NA)	60.9 (NA)	1	1	—	1	—	—	3	3	—	—

Apo, apolipoprotein; CIMT, carotid intima-media thickness; CRP, C-reactive protein; FMD, flow-mediated diameter; HDL, high-density lipoprotein; ICAM-1, intercellular adhesion molecule 1; IL-6, interleukin 6; IQR, interquartile range; LDL, low-density lipoprotein; Lp (a), lipoprotein a; NA, not available; no., number; TC, total cholesterol; TG, triglycerides; TNF- α , tumor necrosis factor α ; VCAM-1, vascular cell adhesion molecule 1.

^aStudy characteristics of HDL and LDL did not vary in comparison to TC, therefore, are not presented in the table.

Outcome	No. of estimates included in analysis	No. of women in intervention arm	No. of women in control arm	Difference, Mean (95% CI)	Quality of evidence
Inflammation markers					
CRP, mg/L	12	640	652	-0.01 (-0.26; 0.23)	B
TNF- α	5	103	79	-0.10 (-0.51; 0.30)	A
IL-6, ng/L	3	70	69	-0.10 (-0.21; 0.02)	A
Haemostatic factors					
Fibrinogen, g/L	5	194	202	-0.17 (-0.36; 0.03)	A
Homocysteine, μ mol/L	5	217	206	-0.17 (-0.63; 0.29)	B
Lipids					
TC, mmol/L	439	1,683	1,705	-0.27 (-0.41; -0.13)	A
HDL, mmol/L	50	1,808	1,724	0.09 (-0.00; 0.18)	A
LDL, mmol/L	51	1,724	1,740	-0.25 (-0.37; -0.13)	B
TG, mmol/L	48	1,669	1,688	-0.20 (-0.28; -0.11)	B
LP(a), g/L	8	421	429	0.22 (-0.15; 0.58)	B
Apo A-1, g/L	17	545	568	0.04 (0.02; 0.07)	A
Apo B, g/L	17	545	568	-0.13 (-0.22; -0.03)	A

*Statistically significant results are bold; Difference in mean is pooled using random effect model

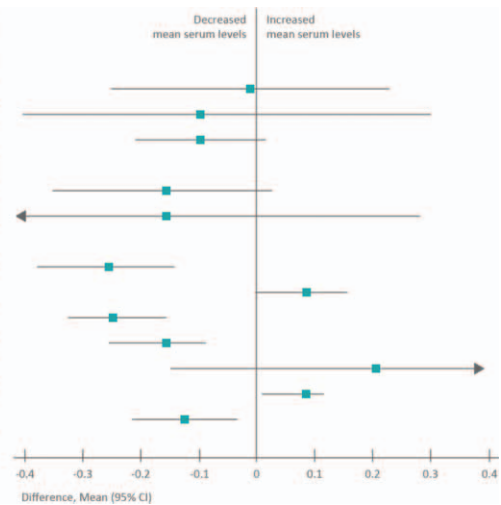


FIG. 2. The associations between phytoestrogen supplementation and inflammation markers, fibrinogen, homocysteine, and blood lipids. Quality of evidence was evaluated using *Grading of Recommendations Assessment, Development and Evaluation* (GRADE): A (High): We are very confident that the true effect lies close to that of the estimate of the effect. Further research is very unlikely to change our confidence in the estimate of effect; B (Moderate): We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different. Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate; C (Low): Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect. Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate; D (Very Low): We have little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect. Apo, apolipoprotein; CRP, C-reactive protein; HDL, high-density lipoprotein; IL-6, interleukin-6; LDL, low-density lipoprotein; LP(a), lipoprotein a; TC, total cholesterol; TG, triglycerides; TNF- α , tumor necrosis factor α .

(Fig. 3, Supplementary Fig. 7, <http://links.lww.com/MENO/A627>).

Oxidative stress

Seven RCTs investigated the effect of lignans^{86,59} and soy-derived phytoestrogens^{54,73,77,87,88} on oxidative stress markers (Supplementary Table 4, <http://links.lww.com/MENO/A627>); yet, we were not able to pool the estimates

because different markers were reported. The protein-rich soy bean intervention resulted in a significant increase in paraoxonase 1 (PON-1) activity as compared with whey protein.⁵⁴ Also, the textured soy protein intervention improved total antioxidant capacity and malondialdehyde compared to no intervention,⁸⁷ whereas no beneficial effects on oxidative stress parameters such as plasma lipid peroxidation, catalase, or glutathione peroxidase were observed after isolated

Outcome	No. of estimates included in analysis	No. of women in intervention arm	No. of women in control arm	Difference, Mean (95% CI)	Quality of evidence
Cell adhesion molecules					
VCAM-1, ng/mL	6	322	321	-3.41 (-16.73; 9.90)	A
ICAM-1, ng/mL	6	322	261	-18.86 (-30.06; -7.65)	A
E-selectin, ng/mL	6	356	355	-2.32 (-4.05; -0.59)	B
Endothelial metabolites					
NOx, μ mol/L	6	287	286	2.67 (-0.03; 5.36)	A
Endothelin-1, pg/mL	4	214	213	0.07 (-0.09; 0.22)	B
Vascular function					
FMD, %	7	288	288	0.28 (-0.68; 1.25)	A
Atherosclerosis					
Mean CIMT, μ m/year	4	389	389	9.34 (0.39; 18.29)	C

*Statistically significant results are bold; Difference in mean is pooled using random effect model

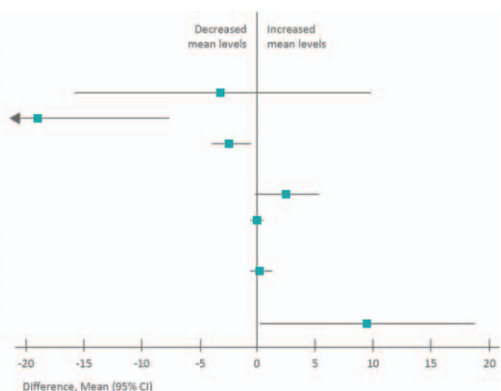


FIG. 3. The associations between phytoestrogen supplementation and cell adhesion molecules, endothelial metabolites, vascular function, and carotid atherosclerosis. Quality of evidence was evaluated using *Grading of Recommendations Assessment, Development, and Evaluation* (GRADE): A (High): We are very confident that the true effect lies close to that of the estimate of the effect; Further research is very unlikely to change our confidence in the estimate of effect; B (Moderate): We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different. Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate; C (Low): Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect. Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate; D (Very Low): We have little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect. CIMT, carotid intima-media thickness; FMD, flow-mediated diameter; ICAM-1, intercellular adhesion molecule 1; NOx, nitric oxide products; VCAM-1, vascular cell adhesion molecule 1.

isoflavones⁷⁷ and after one soy milk intervention.⁷³ Another soy milk intervention resulted in improved total antioxidant capacity.⁸⁸ Lignan-rich sesame seed powder resulted in an increase of serum γ -tocopherol and of the ratio of α - and γ -tocopherol to TC and a decrease in the levels of thiobarbituric acid reactive substances in oxidized LDL as compared to rice powder, whereas the lag time of LDL oxidation did not change after either treatment.⁵⁹

Sensitivity analyses and assessments of bias, study quality, and heterogeneity

There was substantial heterogeneity in quality across the available studies with 26 (46%) and 21 (38%) of the included RCTs demonstrating poor and fair quality, respectively (Supplementary Table 2, <http://links.lww.com/MENO/A627>). Furthermore, Figures 2 to 4 indicate the overall quality of evidence based on the GRADE approach and taking into account the risk of bias, study design, consistency and directness of findings.²⁰ All analyses except a single one (on the association between phytoestrogen supplementation and IL-6) showed high between study heterogeneity; with an I^2 estimate exceeding 75% ($P < 0.001$ for the Cochrane χ^2 statistic) (Supplementary Figs 1-7, <http://links.lww.com/MENO/A627>). The heterogeneity between the investigated parameters was barely explained using “meta regression method.” Factors such as age, time since menopause onset, smoking status, type of phytoestrogen administration, sample size, and study quality are, however, suggested to be the sources of heterogeneity regarding the association of phytoestrogen supplementation with TC, LDL, TG, and Apo A-1. In subgroup analyses, we found significant reductions for TC, TG, and LDL for studies lasting longer than 8 weeks, whereas in shorter trials there was no such effect. Meta-regression analysis is, however, suggesting that this criterion was not the source of heterogeneity between the studies (Supplementary Table 6, <http://links.lww.com/MENO/A627>). Although overall there was no association between phytoestrogen supplementation and serum high-sensitivity CRP, when stratified the analysis by median age of trial participants and study location, in women above the median age of 55.7 years and in studies from the Middle East, phytoestrogen supplementation was associated with a significant decrease in serum hs-CRP. Also, serum Lp(a) increased in women older than 5.3 years since menopause onset (pooled mean difference of changes 0.50 g/L; Supplementary Table 6, <http://links.lww.com/MENO/A627>), whereas in the main analysis we found no association; however, this result was based on estimates from a single RCT.⁵⁷

Leave-one-out sensitivity analysis showed that the pooled estimates were not influenced by any specific study included (Supplementary Figs. 8-15, <http://links.lww.com/MENO/A627>) except in the case of Apo A-1 in which the significant overall estimate might be driven by three RCTs^{36,50,57} indicating no consistency (Supplementary Fig. 14, <http://links.lww.com/MENO/A627>). The funnel plots for the analyses of the association of phytoestrogen supplementation and TG and CRP were asymmetrical, with Egger's P values of 0.04 and

0.002 for CRP and TG, respectively, indicating the presence of publication bias (Supplementary Figs. 16-23, <http://links.lww.com/MENO/A627>). The remaining plots involving a minimum of eight studies remained symmetrical with non-significant Egger's P values ($P > 0.05$).

DISCUSSION

We conducted an extensive systematic review of the benefits of phytoestrogens across a broad range of CVD risk factors in postmenopausal women. Our findings suggest that beneficial effects of phytoestrogens in CVD could be mediated via improvements in some serum lipids and cell adhesion molecules. Regardless of an increase in CIMT with phytoestrogen supplementation, it has to be noted that the observed effect size could be considered small and that the corresponding studies were of low quality. Hence, the evidence provided by these studies probably falls behind the evidence for cardiovascular benefits of phytoestrogens in aging women.

Based on the GRADE approach, there was, in general, high quality of evidence that phytoestrogen supplementation improves serum TC, LDL, TG, Apo A-1, and Apo B. Also, the cholesterol-lowering effect of phytoestrogens was more apparent in postmenopausal women with high initial cholesterol concentrations, indicating that women with dyslipidemia may benefit the most from phytoestrogen supplements. Based on the high quality of the evidence we did not detect an increase of HDL following phytoestrogen supplementation, yet, we found an increase of Apo A-1 which plays a vital role in reverse cholesterol transport and cellular cholesterol homeostasis.⁸⁹ Also, based on moderate-quality evidence phytoestrogens did not decrease serum Lp(a) which is in accordance with a recent meta-analysis on supplemental soy isoflavone intake.¹³ There was moderate and high quality of evidence for improvement in serum E-selectin and ICAM-1, respectively, and high-quality evidence of no effect on VCAM-1. The largest crossover trial found differences in the VCAM-1 response by estrogen receptor (ER) β AluI genotype. There was a significant VCAM-1 response after both isoflavone and placebo treatments in women with the AA genotype, but not with the GG or GA genotypes,⁶⁹ indicating that single-nucleotide polymorphisms in ERs may cause the variability in response to isoflavones.^{90,91}

Although the evidence for a modest disadvantageous effect of phytoestrogens on CIMT was of low quality,^{44,81,82} these results may warrant some attention as they are in line with the so called “timing hypothesis.” This hypothesis suggests that the vasoprotective effects of estradiol may be age dependent and could be lost after a prolonged period of hypoestrogenicity.⁹²⁻⁹⁶ To support this hypothesis: in a single trial phytoestrogen supplementation reduced subclinical atherosclerosis in healthy younger postmenopausal women (median age, 53 years; <5 years postmenopausal) at low CVD risk.⁸² Although two trials included in this analysis were conducted in women who were around 13⁸¹ and 9 years⁴⁴ after menopause onset (and diagnosed with T2D and prehypertension), both reported no effect of phytoestrogen supplements on CIMT. Individuals with

diabetes are, however, at increased risk of developing atherosclerosis, whereas T2D can also alter the expression of ER β (the main phytoestrogen-binding receptor),⁹⁷ which may have also influenced our pooled estimate.

High quality of evidence did not indicate changes in FMD and serum NOx while moderate quality of evidence showed no changes in endothelin-1 with phytoestrogen intervention. Women included in our meta-analysis had relatively high mean baseline FMD ranging between 8.6% and 13.7%, and although previous evidence suggested that isoflavones may be beneficial in women with decreased FMD only⁹⁸ due to the small number of trials reporting on FMD we were not able to stratify the analyses by median baseline FMD. Similarly, due to the limited number of trials we were not able to further explore the null findings observed in serum NOx and endothelin-1.

There was moderate quality of evidence that phytoestrogen supplementation did not change inflammatory markers and homocysteine and high-quality evidence that there was no effect on fibrinogen levels. Yet, in subgroup analyses, serum CRP decreased in older women (above median age of 55.7 years) and in trials conducted in the Middle East. It may be that phytoestrogen supplementation is beneficial in elderly women who usually have higher CRP levels⁹⁹ and that a prolonged exposure to phytoestrogens (such as in areas where phytoestrogens are a regular part of the diet) may play an important role in modifying this association. In line with our hypothesis, a previous meta-analysis indicated a beneficial role of soy isoflavones in decreasing CRP in women with elevated baseline CRP levels,¹¹ whereas in our study only 3 out of 11 RCTs included women with elevated serum CRP levels (mean >3 mg/L).

The beneficial role of phytoestrogen supplementation on markers of oxidative stress was inconsistent and the quality of evidence was only moderate. Yet, *in vitro* and animal studies identified various mechanisms supporting a role of phytoestrogens in improving antioxidant status mostly by their action as free radical scavengers¹⁰⁰ as well as by increasing mitochondrial glutathione and gene expression of antioxidative enzymes.^{101,102} Thus oxidative stress remains an important factor contributing to increased CVD risk in aging women and merits further investigation.

Strengths and weaknesses

This review was conducted in accordance with the Cochrane guidelines¹⁰³ and we used the Cochrane risk of bias tool¹⁰⁴ and the GRADE¹⁰⁴ approach to rate the quality of the evidence. In order to identify as many relevant studies as possible and reduce the risk of publication bias, a highly sensitive search strategy was used and additional resources were searched including the reference list of included trials and relevant systematic reviews. Nevertheless, this study has a number of limitations. First, although conventional funnel plots and Egger test estimates indicated only a minimal publication bias, these methods are limited by their qualitative nature. We can, therefore, not exclude that measured or unmeasured publication bias limits

our findings. Second, study quality, women's age, time since menopause onset, and smoking status contributed to the heterogeneity of findings. Yet, the number of available studies in some analyses was small ($n \leq 3$), precluding our ability to investigate the sources of the observed heterogeneity. Furthermore, the supplements provided to trial participants may vary in quality and composition; also only four RCTs matching our search criteria were found after 2015. Moreover, the formulation and quality of newer supplements may differ as compared to the supplements used ≥ 4 years ago. The capacity of individuals to produce equol (by gut microbiome) may be one of the most important determinants of phytoestrogen effectiveness.¹⁰⁵ For example, Asian individuals have greater ability than non-Asians to produce equol (a metabolite of isoflavone called daidzein).¹⁰⁶ Also, emerging evidence has suggested an association between equol-producer status, genetic factors (ER polymorphism), and vascular function.¹⁰⁷ The existing trials did not properly address this issue; thus, it is necessary that future trials investigate how phytoestrogen metabolites in serum/urine (as a proxy of phytoestrogen bioactivity) contribute to any effects of phytoestrogens on women's health. Also, the effect of phytoestrogen supplementation may depend on the level of dietary intake: The mean intake of isoflavones in postmenopausal women may vary from 0.779 mg/d in USA¹⁰⁸ to 11.3 to 41.3 mg/d in Japan.¹⁰⁹ Although we used the study location as an indicator of dietary habits, a comprehensive assessment of women's dietary habits is necessary to distinguish women who are regularly consuming phytoestrogen rich food. Lastly, our study did not include the major CVD endpoints because corresponding RCTs on clinical CVD endpoints are not available. The included outcomes (ie, CRP, homocysteine, serum lipids, CIMT) may, however, be considered as proxies of CVD risk.¹¹⁰⁻¹¹² Nevertheless, the risk of a fatal CVD event depends on several influences including genetic factors and lifestyle behaviors, which can strongly increase or reduce an individual's cardiovascular risk.¹¹³

Our search identified three reviews of clinical trials conducted in postmenopausal women that reported improvements in blood lipids¹¹⁴ with flaxseed and improvements in serum homocysteine, lipids,¹⁴ and FMD⁹⁸ with isoflavone supplementation. This meta-analysis strictly included trials that studied phytoestrogen interventions alone in comparison with controls, whereas the earlier meta-analyses also included trials with combined interventions (eg, soy isoflavones and low-fat diet). Thus, those earlier studies left unclear whether the beneficial effects observed in these studies were enhanced by beneficial lifestyle changes or whether they can be attributed to flaxseed and isoflavones alone. Also, those reviews did not use the GRADE approach to rate the quality of the evidence and therefore the strength of the evidence published previously was unclear.

Implications for future research

In Figure 4 we present the overall findings, quality of evidence, and factors which may have influenced the observed associations. Herewith we provide hints for the

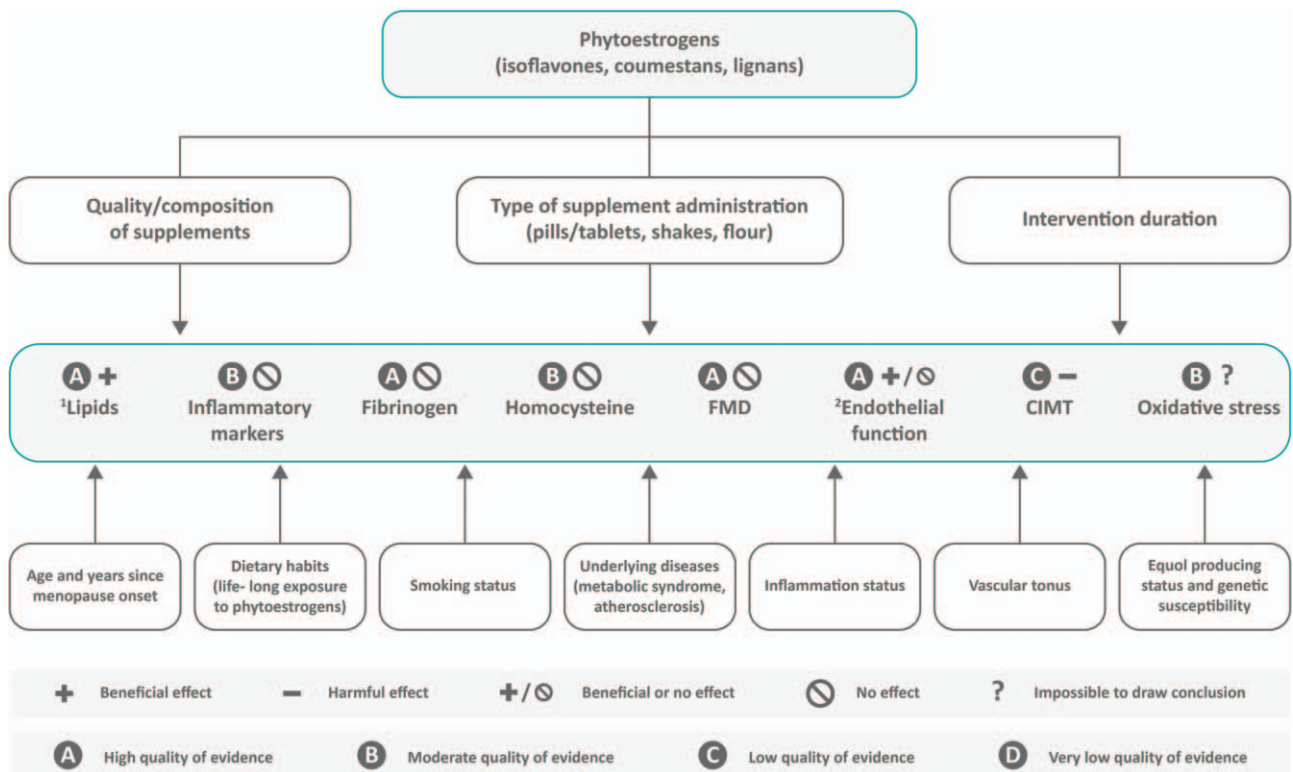


FIG. 4. Illustrative summary of overall findings. Quality of evidence was evaluated using *Grading of Recommendations Assessment, Development and Evaluation* (GRADE): A (High): We are very confident that the true effect lies close to that of the estimate of the effect; Further research is very unlikely to change our confidence in the estimate of effect; B (Moderate): We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different. Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate; C (Low): Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect. Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate; D (Very Low): We have little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect. ¹The beneficial effect for lipids refers to total cholesterol, low-density lipoprotein, triglycerides, apolipoprotein A-1, apolipoprotein B; ²The beneficial effect for endothelial function refers to intercellular adhesion molecule 1 and E-selectin. CIMT, carotid intima-media thickness; FMD, flow-mediated diameter.

directions future research may take. In particular, to study the effects of phytoestrogens on cardiovascular health in aging women, well-designed clinical trials should (i) compare the effectiveness of three major types of phytoestrogens (isoflavones, lignans, coumestans) with placebo for a sufficiently long duration of at least 1 year in healthy and in women with impaired cardiometabolic health; (ii) consider a broad spectrum of intermediate CVD risk factors; (iii) evaluate the ER polymorphism, equol producing status, and measure blood and urine phytoestrogen metabolites as a proxy of their bioavailability; and (iv) carefully examine the dietary habits of postmenopausal women at baseline and during the follow-up.

CONCLUSIONS

This meta-analysis of clinical trials suggests that phytoestrogen supplementation improves the CVD risk profile in postmenopausal women, particularly by beneficially influencing blood lipids and some parameters of endothelial function while such association could not be observed for plasma fibrinogen and FMD. Although modest, a deleterious effect on CIMT progression may, however, be present in particular in

postmenopausal women at increased risk of developing atherosclerosis. Because of the limited quality of the evidence, we cannot draw firm conclusions on how phytoestrogens may affect inflammatory markers, homocysteine, and oxidative stress. Therefore, future rigorous clinical trials are needed to further explore the potential of phytoestrogens in improving menopausal health.

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